Object Follow (No-PTZ)

Function package: `microros_v2_kcf_follow_pkg`

1. Function Description

`microros_v2_kcf_follow_pkg` is a following controller based on the target bounding box. It only subscribes to the target box published by the tracking node, and then controls the PTZ and/or body motion.

Therefore, the complete functionality consists of two parts:

1. The tracker is responsible for obtaining the target bounding box from the image, for example `smart_tracker/super_tracker` or the KCF tracking node.
2. `tracking_bbox_follow_node` computes the PTZ angle, body angular velocity, and body linear velocity from the target box center and target box size.

The default control rate is 10 Hz. When the target box is invalid, the confidence is too low, or it has not been updated for more than `target_stale_timeout_sec`, the node immediately publishes zero velocity; the source code has no "allow continued motion for 5 frames" lost-target tolerance logic.

2. Control Chain

```
Camera/visual tracker
    └── /tracking_bbox
          └── tracking_bbox_follow_node (10 Hz)
                ├── PTZ following: target center -> /cmd_ptz_angle
                ├── Body turning: target horizontal error or PTZ deflection ->
/cmd_vel.angular.z
                └── Distance following: target box size error ->
/cmd_vel.linear.x
```

PTZ Version

By default `enable_ptz_follow:=true`:

- Uses independent X/Y PD control to update the PTZ angle from the pixel error of the target center relative to the image center.
- After the PTZ horizontal angle deviates from the initial angle by `servo1_engage_deg`, body PD control is used for `angular.z`, letting the body gradually follow the PTZ back to center.
- At the same time, `linear.x` can be controlled based on the target box size.

No-PTZ Version

A fixed camera cannot adjust its line of sight through the PTZ; the following must be set at startup:

```
enable_ptz_follow:=false publish_initial_ptz_command:=false
```

In this case, the node uses the horizontal pixel error of the target center to control the body `angular.z`, and controls `linear.x` based on the target box size. Setting only `publish_initial_ptz_command:=false` does not disable the PTZ following logic, so it cannot be the only modification for a No-PTZ configuration.

3. Topics and Messages

Topic	Direction	Message Type	QoS	Description
`/tracking_bbox`	Subscribe	Auto-detected by default	`BEST_EFFORT`, depth=1	Target bounding box input
`/cmd_vel`	Publish	`geometry_msgs/msg/Twist`	`BEST_EFFORT`, depth=1	`linear.x` forward/backward, `angular.z` left/right turn
`/cmd_ptz_angle`	Publish	`geometry_msgs/msg/Vector3`	depth=10	Published in PTZ mode; `x` is the horizontal angle and `y` is the pitch angle, in degrees
`/JoyState`	Subscribe	`std_msgs/msg/Bool`	depth=10	When `true`, pauses automatic following and publishes zero velocity

The node is also compatible with:

- `smart_tracker/msg/TrackBox`, auto-detected via `tracking_bbox_msg_type:=auto`;
- `std_msgs/msg/Int32MultiArray`: when the length is at least 11, it is parsed as `[valid, x1, y1, x2, y2, cx, cy, width, height, image_width, image_height]`; when the length is 4 to 10, the first four elements are taken as the bounding box;
- messages with `x1`, `y1`, `x2`, `y2` fields.

The bounding box format is uniformly the top-left corner `(x1, y1)` and the bottom-right corner `(x2, y2)`. The node automatically sorts the coordinates and checks the width, height, and confidence.

4. Startup Steps

The following commands assume the workspace is `~/microros_car_v2_ws` and the ROS 2 environment has been configured.

4.1 Start the Chassis, Agent, Camera, and Model

Terminal 1:

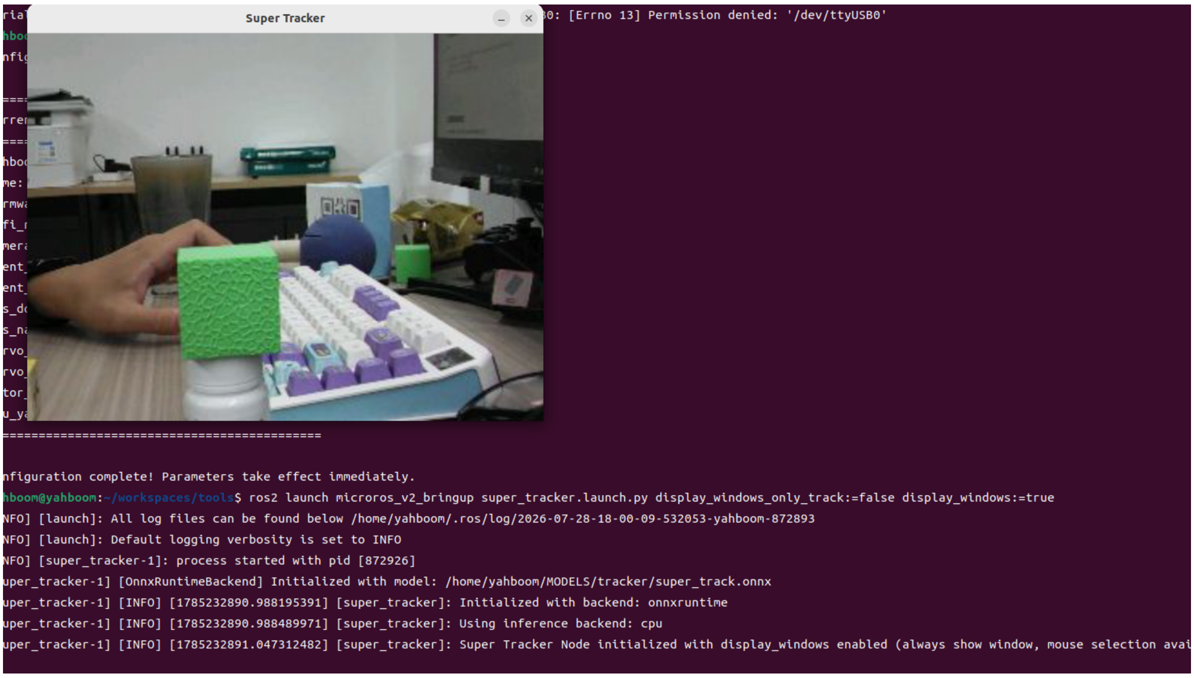
```
ros2 launch microros_v2_bringup car_base.launch.py
```

```
yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-15-04-46-269642-yahboom-579370
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [579411]
[INFO] [camera_driver-2]: process started with pid [579413]
[INFO] [robot_state_publisher-3]: process started with pid [579415]
[INFO] [joint_state_publisher-4]: process started with pid [579417]
[INFO] [ekf_node-5]: process started with pid [579419]
[micro_ros_agent-1] [1785222287.404236] info     | UDPv4AgentLinux.cpp | init                     | running...             | port: 8090
[robot_state_publisher-3] [INFO] [1785222288.847816851] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785222288.848013042] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785222288.848028942] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848037122] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785222288.848043874] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785222288.848050538] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785222288.848057264] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785222288.848063747] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785222288.848070240] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848076888] [robot_state_publisher]: got segment rfwheel_Link
[joint_state_publisher-4] [INFO] [1785222289.372349515] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[camera_driver-2] [INFO] [1785222289.716161193] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[camera_driver-2] [WARN] [1785222292.705741414] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785222294.072701944] [esp32_ip_camera_publisher]: IP camera stream opened successfully
```

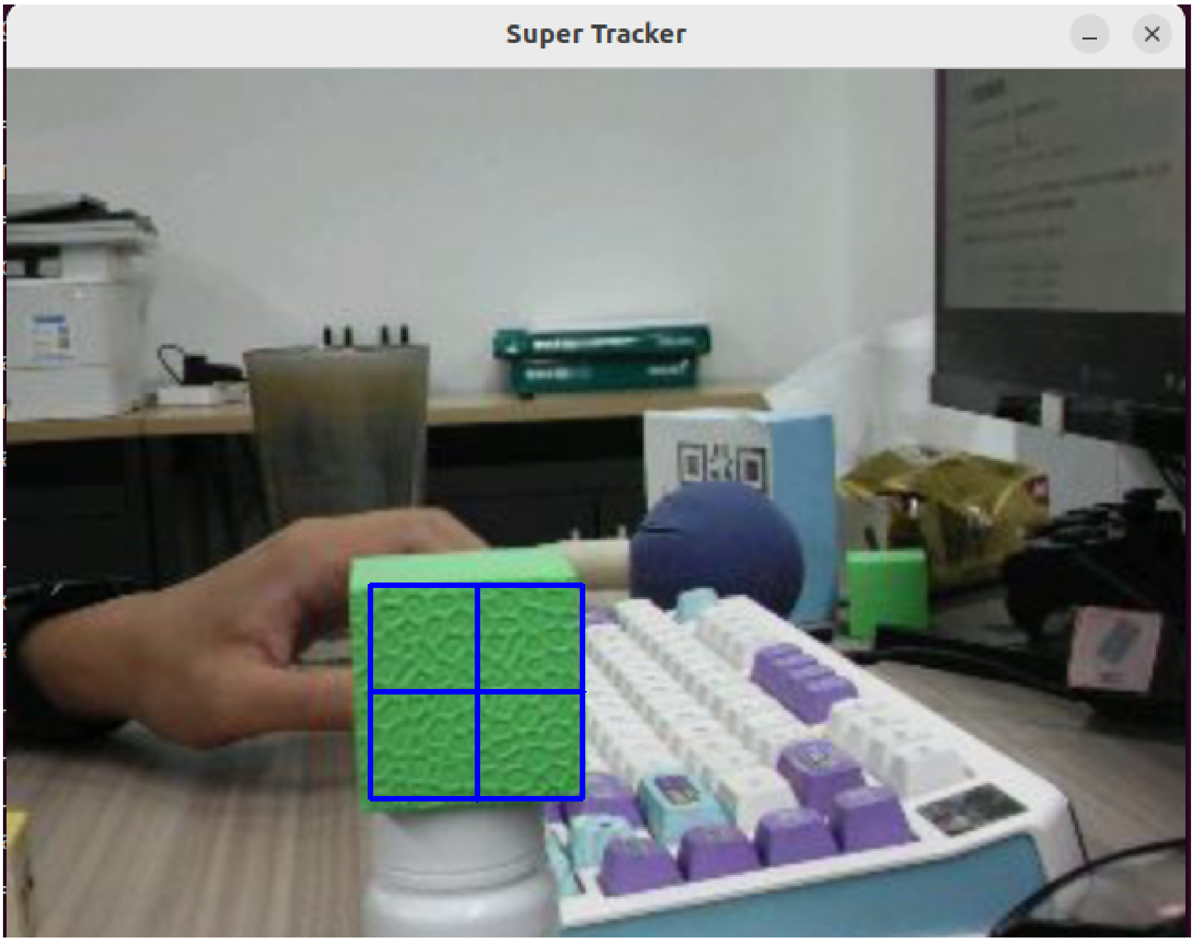
4.2 Start the Target Tracker

Run the program `smart_tracker/super_tracker`, Terminal 2:

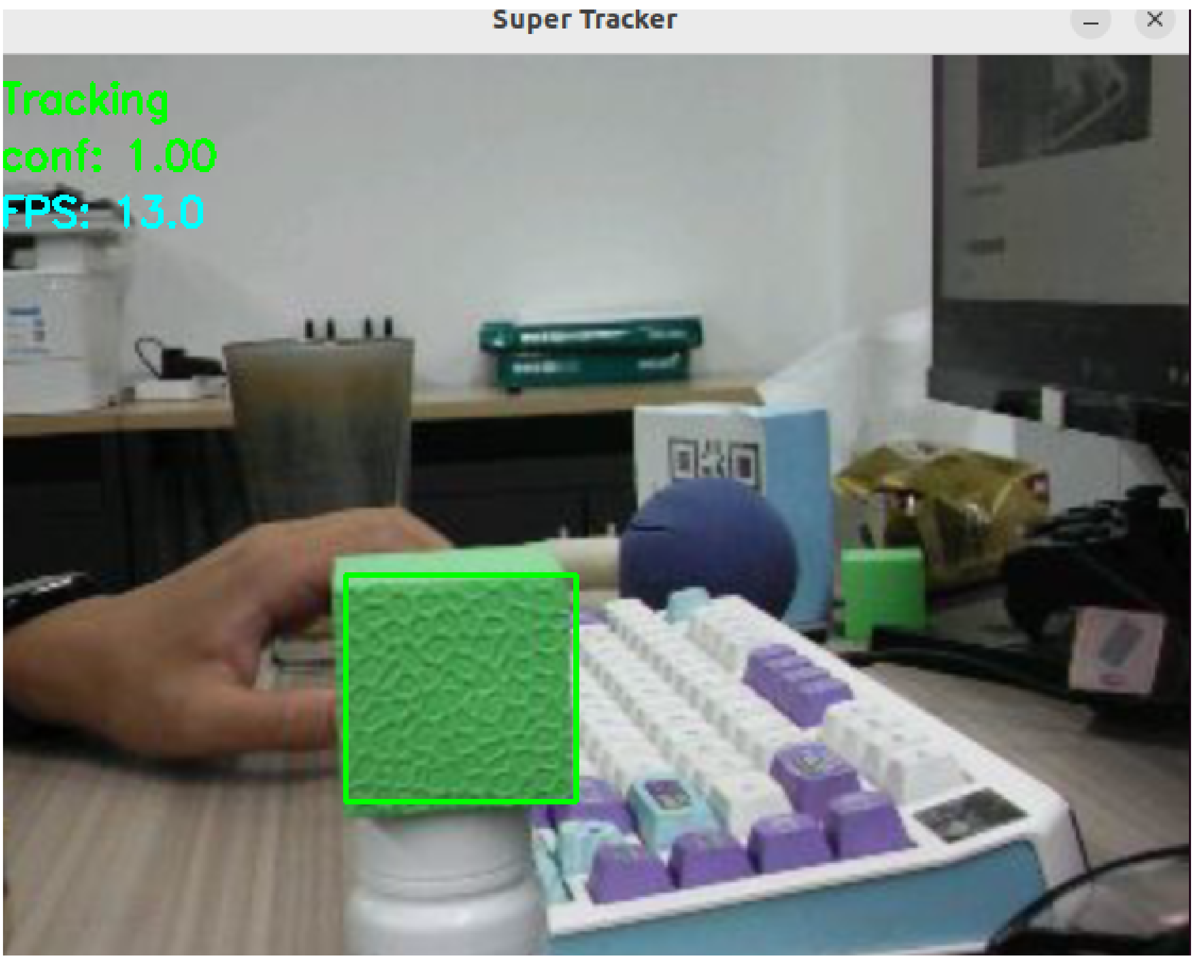
```
ros2 launch microros_v2_bringup super_tracker.launch.py
display_windows_only_track:=false display_windows:=true
```



Press `c` to enter box-selection mode, then use the mouse to draw a box



Then press Enter to start following the coordinate data.

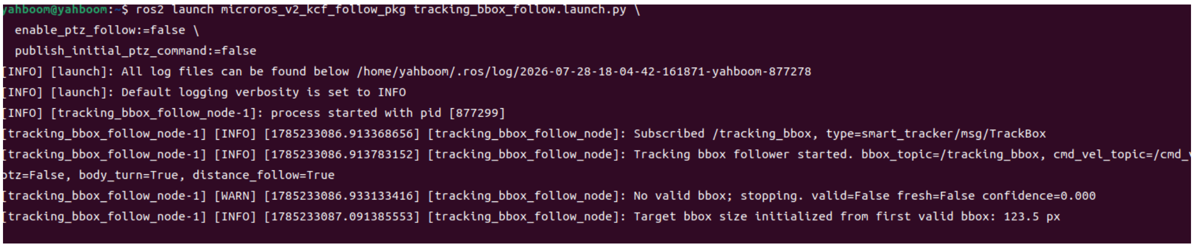


4.4 Start the Follow Controller

Terminal 3:

```
ros2 launch microros_v2_kcf_follow_pkg tracking_bbox_follow.launch.py \
  enable_ptz_follow:=false \
  publish_initial_ptz_command:=false
```

After a successful startup, slowly move the object to test left/right tracking and forward/backward following.



5. Distance Following Algorithm

The node does not estimate distance from the color radius, but instead uses the geometric size of the bounding box:

```
current_size = sqrt(width * height)
size_error = (target_size - current_size) / target_size
```

By default `auto_set_target_bbox_size:=true`. The first valid bounding box received after startup is used as the target distance reference. You can also manually specify `target_bbox_size_px`, in which case the node no longer waits for the first bounding box to initialize the reference.

The distance control uses the PID form; with the default `distance_ki:=0.0`, it is effectively PD:

```
linear.x = linear_dir * (distance_kp * size_error
                       + distance_ki * integral
                       + distance_kd * d_error)
```

- When the target box is smaller than the reference, `size_error > 0`, and it moves forward to approach by default;
- When the target box is larger than the reference, `size_error < 0`, and it moves backward to retreat by default;
- When `abs(size_error) <= distance_tolerance_ratio`, it stops moving forward/backward;
- `linear.x` is clamped to `[-max_linear_x, max_linear_x]`, with a default maximum absolute value of `0.12 m/s`.

6. Key Parameters

General Parameters

Parameter	Default	Description
`tracking_bbox_topic`	`/tracking_bbox`	Target box input topic; the node must be restarted after changing
`tracking_bbox_msg_type`	`auto`	Auto-detect the message type; you can also fill in the full ROS type name
`cmd_vel_topic`	`/cmd_vel`	Body velocity output topic; the node must be restarted after changing
`ptz_command_topic`	`/cmd_ptz_angle`	PTZ angle output topic; the node must be restarted after changing
`joy_state_topic`	`/JoyState`	Joystick override state topic; the node must be restarted after changing
`enable_follow`	`true`	Master follow switch; when disabled, continuously publishes zero velocity
`enable_ptz_follow`	`true`	Enable PTZ following; must be set to `false` for No-PTZ
`enable_body_turn_follow`	`true`	Enable body horizontal turning
`enable_distance_follow`	`true`	Enable forward/backward distance following
`enable_joy_override`	`true`	Pause automatic following when the joystick takes over
`image_width` / `image_height`	`640` / `480`	Used when the input message does not provide the image size
`tracking_confidence_threshold`	`0.1`	Minimum valid confidence
`target_stale_timeout_sec`	`1.2`	Stops the car when the target box has not been updated for longer than this time
`follow_rate_hz`	`10.0`	Control timer rate, minimum 1 Hz

PTZ and Body Turning Parameters

Parameter	Default	Description
`pixel_tolerance`	`15.0` px	PTZ horizontal/pitch pixel deadband
`kp_x` / `kd_x`	`0.6` / `0.15`	PTZ horizontal PD coefficients
`kp_y` / `kd_y`	`0.6` / `0.15`	PTZ pitch PD coefficients
`max_step_xy`	`5.0` deg	Maximum PTZ angle step per control cycle
`horizontal_min_deg` / `horizontal_max_deg`	`-90` / `90` deg	Horizontal angle limits
`pitch_min_deg` / `pitch_max_deg`	`-90` / `20` deg	Pitch angle limits
`initial_horizontal_deg`	`0.0` deg	Initial horizontal angle, also the reference angle for the body to return to center
`initial_pitch_deg`	`-45.0` deg	Initial pitch angle
`body_kp` / `body_kd`	`0.03` / `0.0`	Body turning PD coefficients in PTZ mode
`servo1_engage_deg`	`5.0` deg	Body turning starts when the PTZ horizontal deflection reaches this value
`body_tolerance_deg`	`3.0` deg	Deadband for stopping the body when it returns to center
`max_angular_z`	`0.6` rad/s	Maximum absolute body angular velocity
`body_ff_gain`	`1.0`	Feed-forward coefficient in PTZ mode that compensates the PTZ horizontal angle based on body angular velocity
`fixed_camera_pixel_tolerance`	`15.0` px	No-PTZ body turning stop deadband
`fixed_camera_engage_px`	`30.0` px	No-PTZ body turning activation threshold
`fixed_camera_body_kp` / `fixed_camera_body_kd`	`0.6` / `0.15`	No-PTZ body turning PD coefficients
`body_dir` / `fixed_camera_body_dir`	`-1.0` / `-1.0`	Body turning direction correction; change to `1.0` if necessary
`sign_x` / `sign_y`	`1.0` / `-1.0`	PTZ horizontal/pitch direction correction

Distance Following Parameters

Parameter	Default	Description
`target_bbox_size_px`	`0.0`	Target reference size; when 0, it is automatically set from the first valid box
`auto_set_target_bbox_size`	`true`	Whether to initialize the reference size from the first valid box
`distance_tolerance_ratio`	`0.06`	Relative size deadband, default 6%
`distance_kp` / `distance_ki` / `distance_kd`	`0.18` / `0.0` / `0.02`	Distance PID coefficients
`distance_integral_limit`	`0.5`	Integral limit
`max_linear_x`	`0.12` m/s	Maximum absolute linear velocity
`linear_dir`	`1.0`	Forward/backward direction correction; change to `-1.0` if necessary